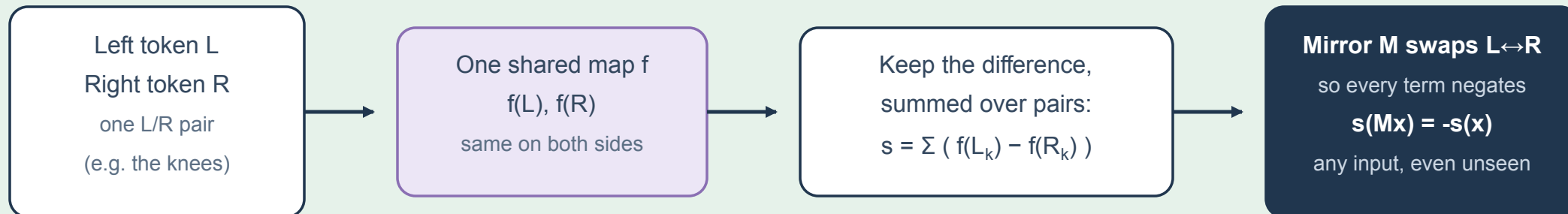


Emergent vs built-in: two ways to get the mirror rule

Two ways a model could satisfy the mirror rule $s(Mx) = -s(x)$: build it in by wiring (top), or hope training teaches it (bottom).

Built-in: shared map, keep only the difference

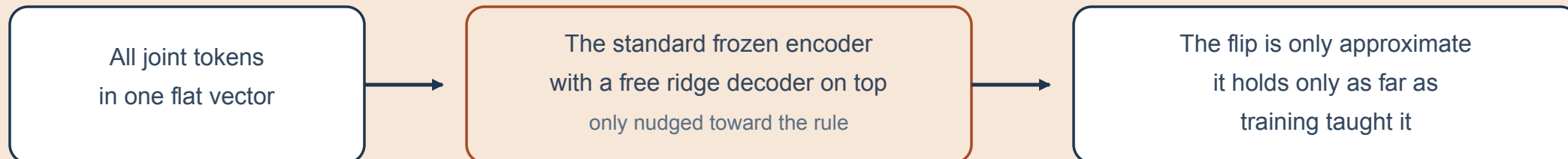
exact: every input, even unseen



mirror slope = **-1.000** for ANY weights (frame averaging over $G=\{I,M\}$)

Emergent hope: one flat vector, nudged by training

approximate: only what training saw



Reflection augmentation is OFF in canonical training (flip probability 0.0): this is a genuine test of the property, not something taught.

Illustrative schematic. Folder labels are dataset annotations, not diagnoses.